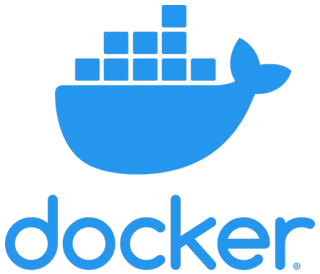
Containers on AWS

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

What is Docker?

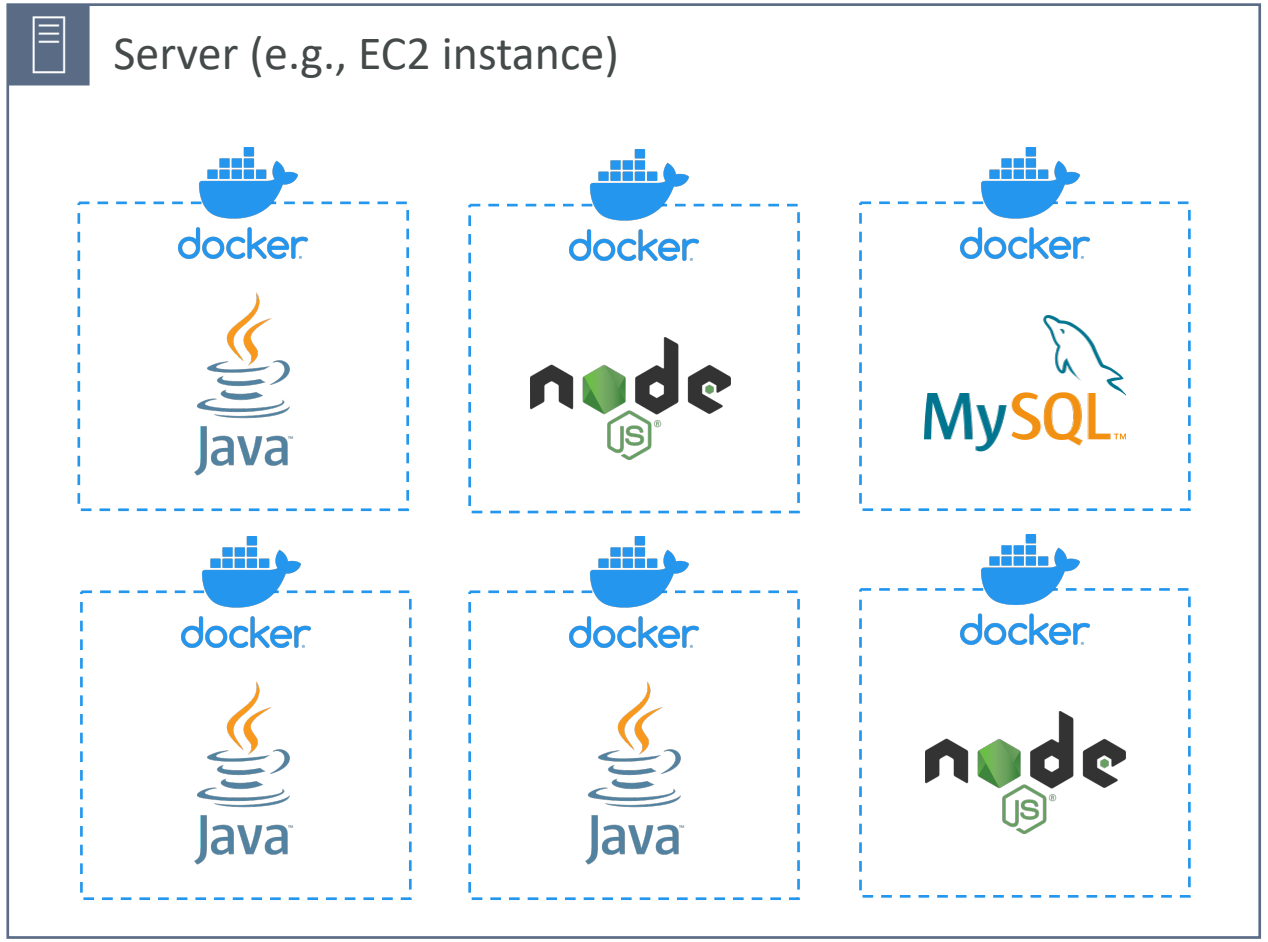


- Docker is a software development platform to deploy apps
- Apps are packaged in **containers** that can be run on any OS
- Apps run the same, regardless of where they're run
  - Any machine
  - No compatibility issues
  - Predictable behavior
  - Less work
  - Easier to maintain and deploy
  - Works with any language, any OS, any technology
- Use cases: microservices architecture, lift-and-shift apps from on-premises to the AWS cloud, …

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Docker on an OS



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Where are Docker images stored?

- Docker images are stored in Docker Repositories

- Docker Hub (https://hub.docker.com)
  - **Public** repository
  - Find base images for many technologies or OS (e.g., Ubuntu, MySQL, …)

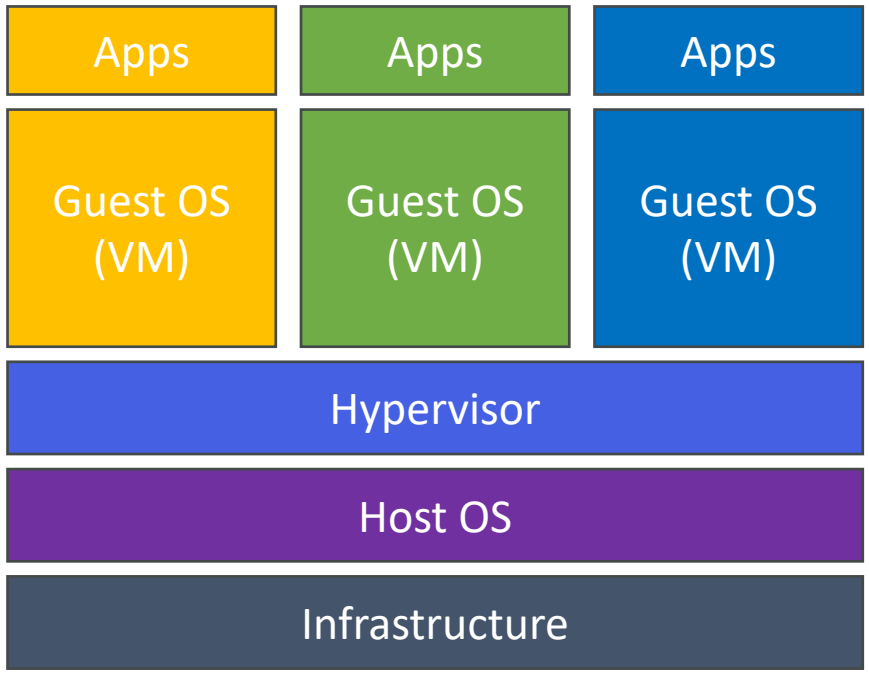
- Amazon ECR (Amazon Elastic Container Registry)
  - **Private** repository
  - **Public** repository (**Amazon ECR Public Gallery** https://gallery.ecr.aws)

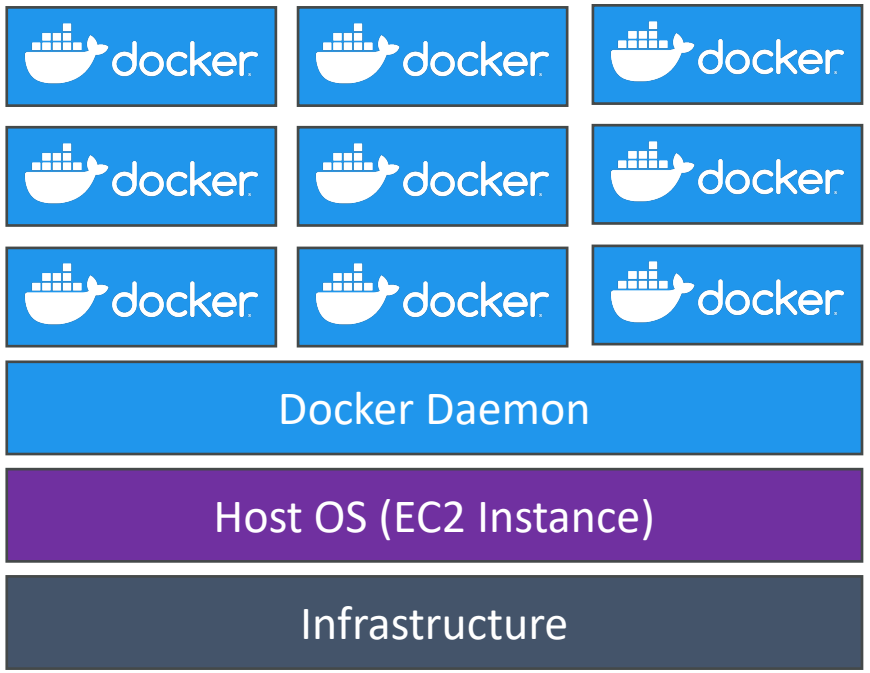
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Docker vs. Virtual Machines

- Docker is "sort of" a virtualization technology, but not exactly
- Resources are shared with the host => many containers on one server

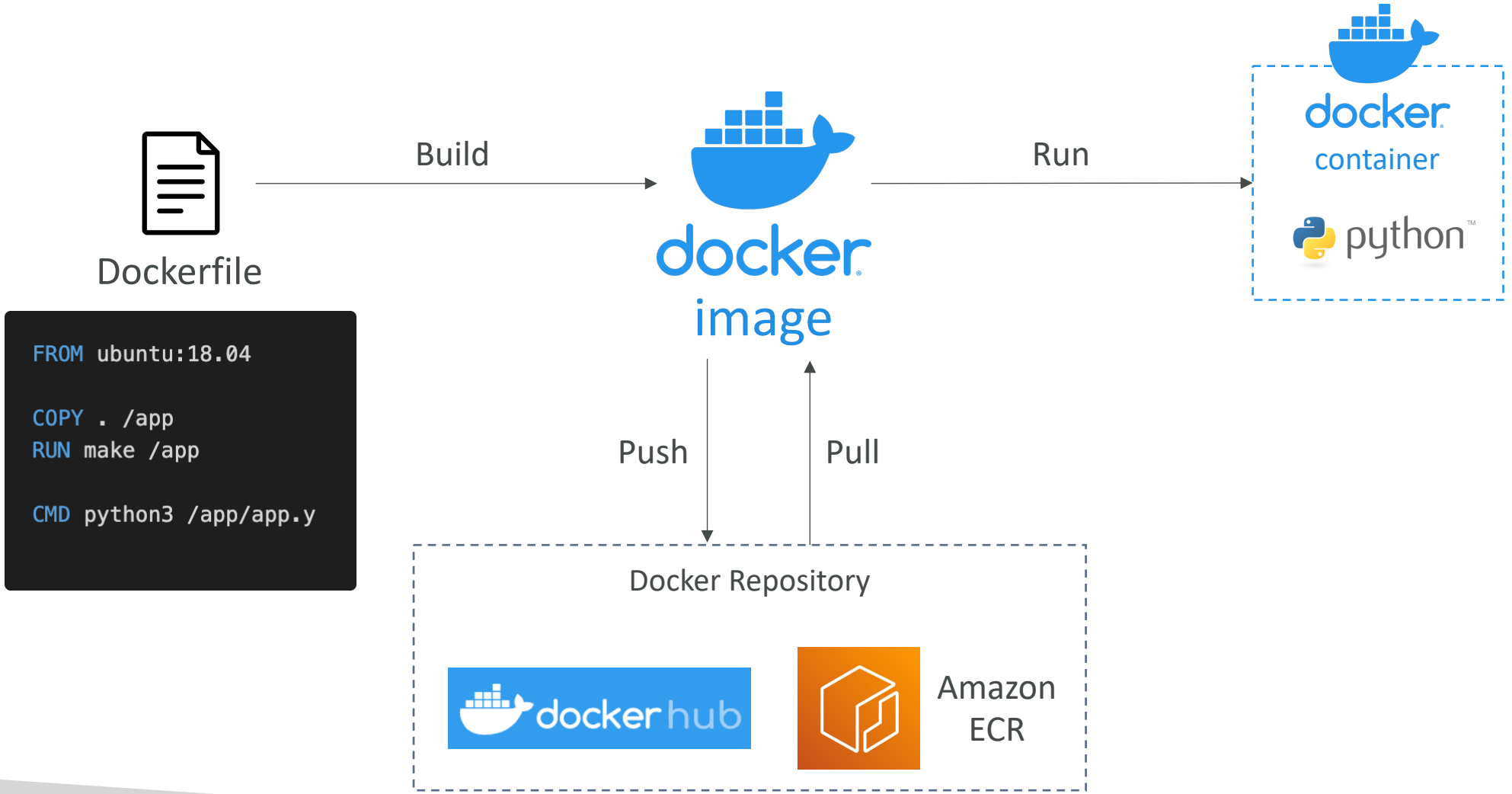




© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Getting Started with Docker



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Docker Containers Management on AWS

- Amazon Elastic Container Service (Amazon ECS)
  - Amazon's own container platform
- Amazon Elastic Kubernetes Service (Amazon EKS)
  - Amazon's managed Kubernetes (open source)
- AWS Fargate
  - Amazon's own Serverless container platform
  - Works with ECS and with EKS
- Amazon ECR:
  - Store container images



Amazon ECS



Amazon EKS



AWS Fargate



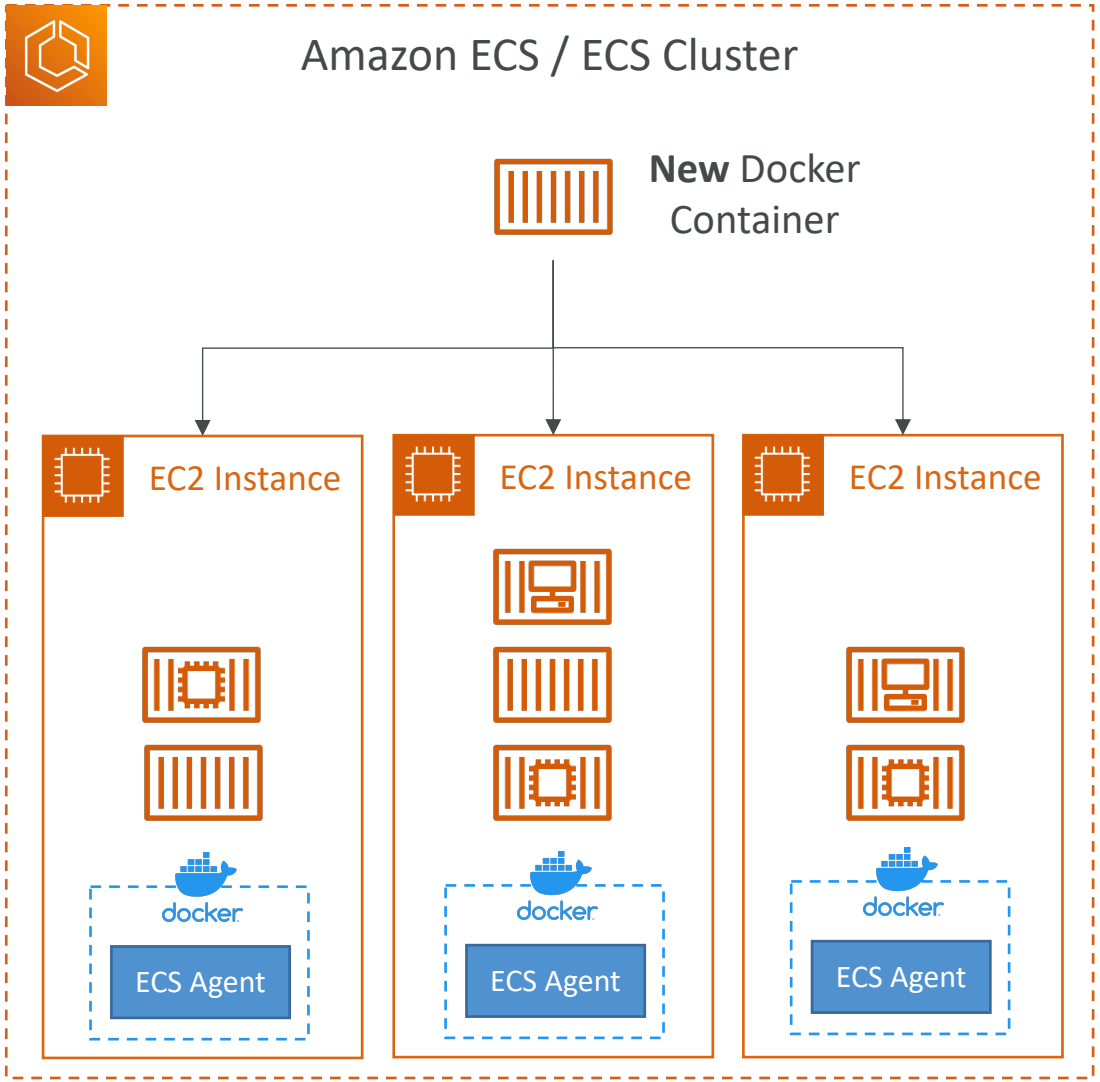
Amazon ECR

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon ECS - EC2 Launch Type

- ECS = Elastic Container Service
- Launch Docker containers on AWS = Launch **ECS Tasks** on ECS Clusters
- **EC2 Launch Type: you must provision & maintain the infrastructure (the EC2 instances)**
- Each EC2 Instance must run the ECS Agent to register in the ECS Cluster
- AWS takes care of starting / stopping containers

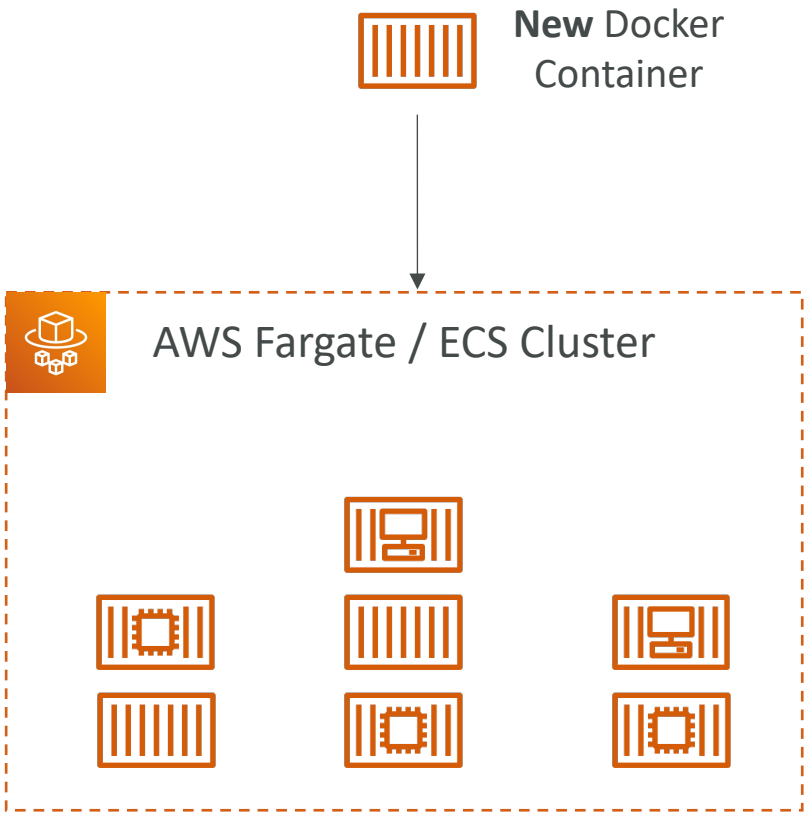


NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Amazon ECS – Fargate Launch Type

- Launch Docker containers on AWS
- **You do not provision the infrastructure (no EC2 instances to manage)**
- **It's all Serverless!**
- You just create task definitions
- AWS just runs ECS Tasks for you based on the CPU / RAM you need
- To scale, just increase the number of tasks. Simple - no more EC2 instances



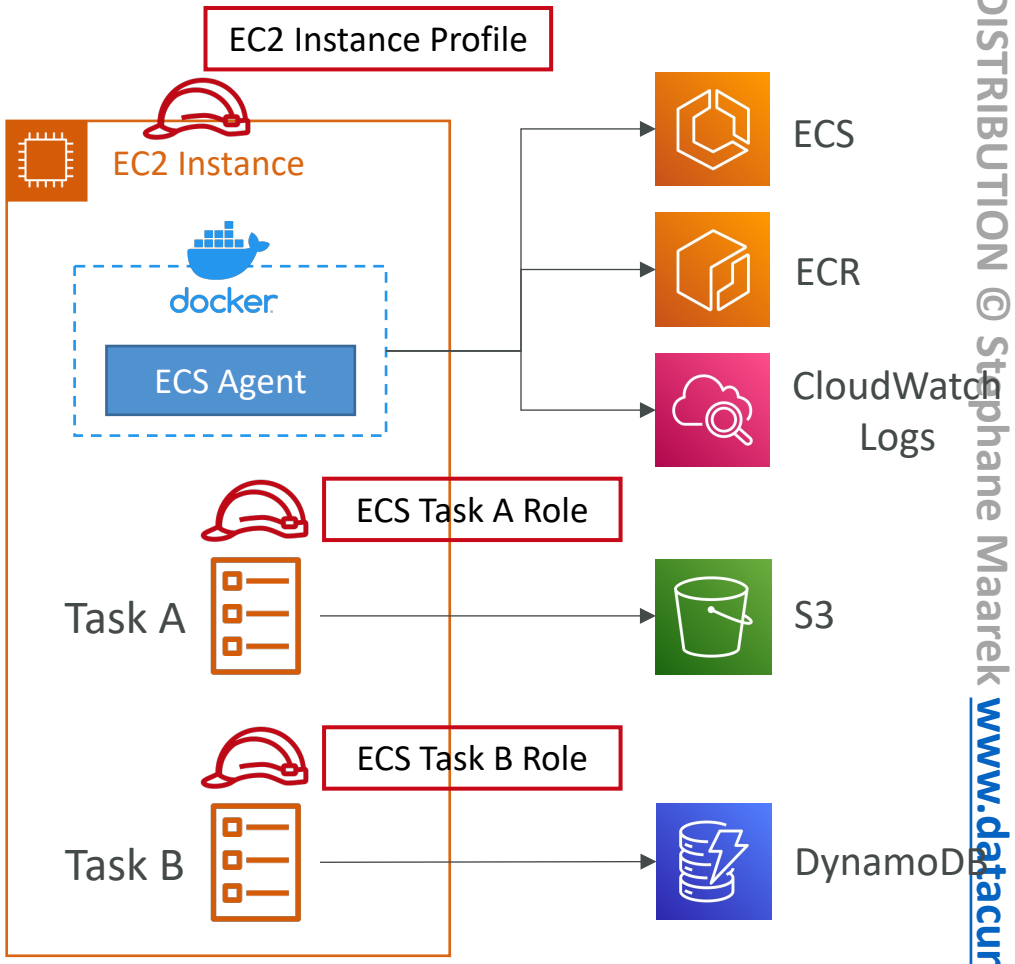
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon ECS – IAM Roles for ECS

- EC2 Instance Profile (EC2 Launch Type only):
  - Used by the ECS agent
  - Makes API calls to ECS service
  - Send container logs to CloudWatch Logs
  - Pull Docker image from ECR
  - Reference sensitive data in Secrets Manager or SSM Parameter Store

- ECS Task Role:
  - Allows each task to have a specific role
  - Use different roles for the different ECS Services you run
  - Task Role is defined in the task definition

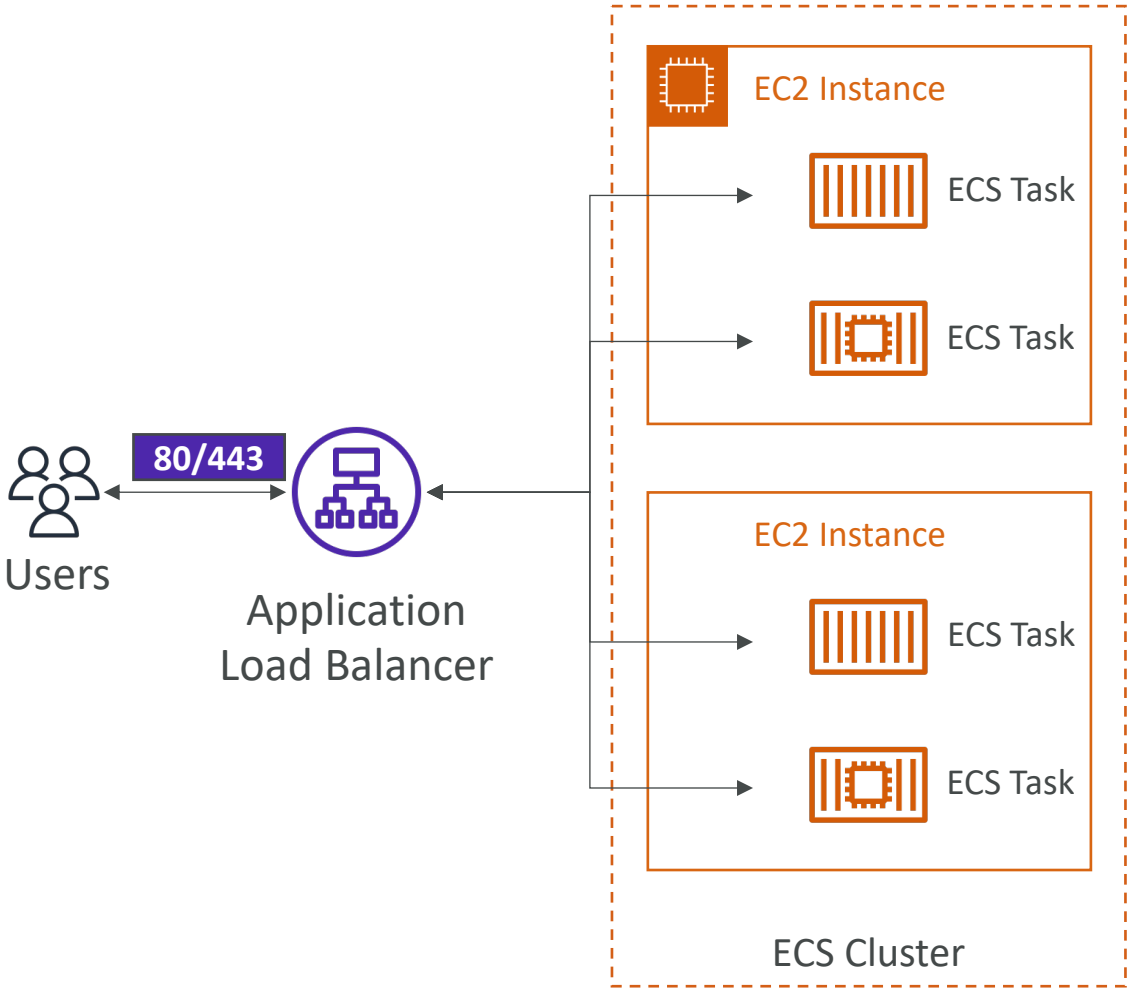


NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek

Amazon ECS – Load Balancer Integrations

- Application Load Balancer supported and works for most use cases
- Network Load Balancer recommended only for high throughput / high performance use cases, or to pair it with AWS Private Link
- Classic Load Balancer supported but not recommended (no advanced features – no Fargate)



© Stephane Maarek

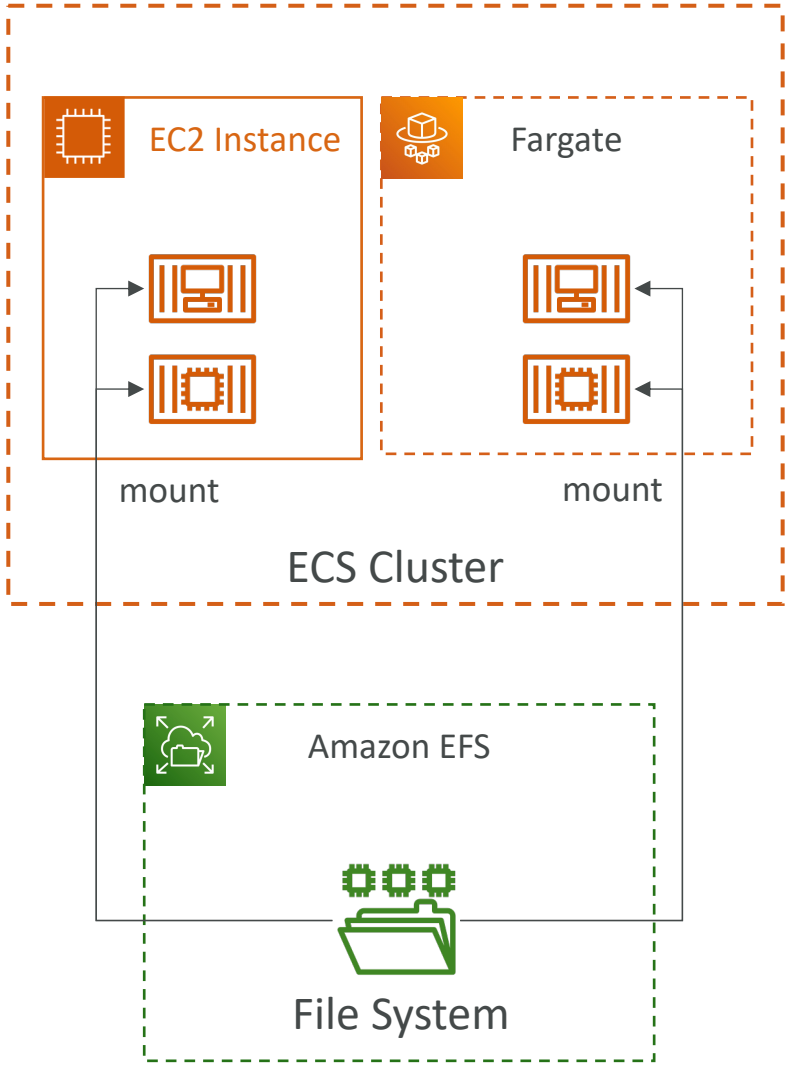
NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon ECS – Data Volumes (EFS)

- Mount EFS file systems onto ECS tasks
- Works for both **EC2** and **Fargate** launch types
- Tasks running in any AZ will share the same data in the EFS file system
- **Fargate + EFS = Serverless**

- Use cases: persistent multi-AZ shared storage for your containers

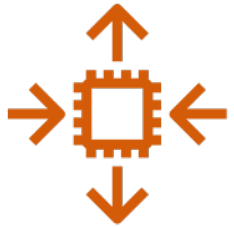
- Note:
  - Amazon S3 cannot be mounted as a file system



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

ECS Service Auto Scaling



- Automatically increase/decrease the desired number of ECS tasks

- Amazon ECS Auto Scaling uses **AWS Application Auto Scaling**
  - ECS Service Average CPU Utilization
  - ECS Service Average Memory Utilization - Scale on RAM
  - ALB Request Count Per Target – metric coming from the ALB

- **Target Tracking** – scale based on target value for a specific CloudWatch metric
- **Step Scaling** – scale based on a specified CloudWatch Alarm
- **Scheduled Scaling** – scale based on a specified date/time (predictable changes)

- ECS Service Auto Scaling (task level) ≠ EC2 Auto Scaling (EC2 instance level)
- Fargate Auto Scaling is much easier to setup (because **Serverless**)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

EC2 Launch Type – Auto Scaling EC2 Instances

- Accommodate ECS Service Scaling by adding underlying EC2 Instances

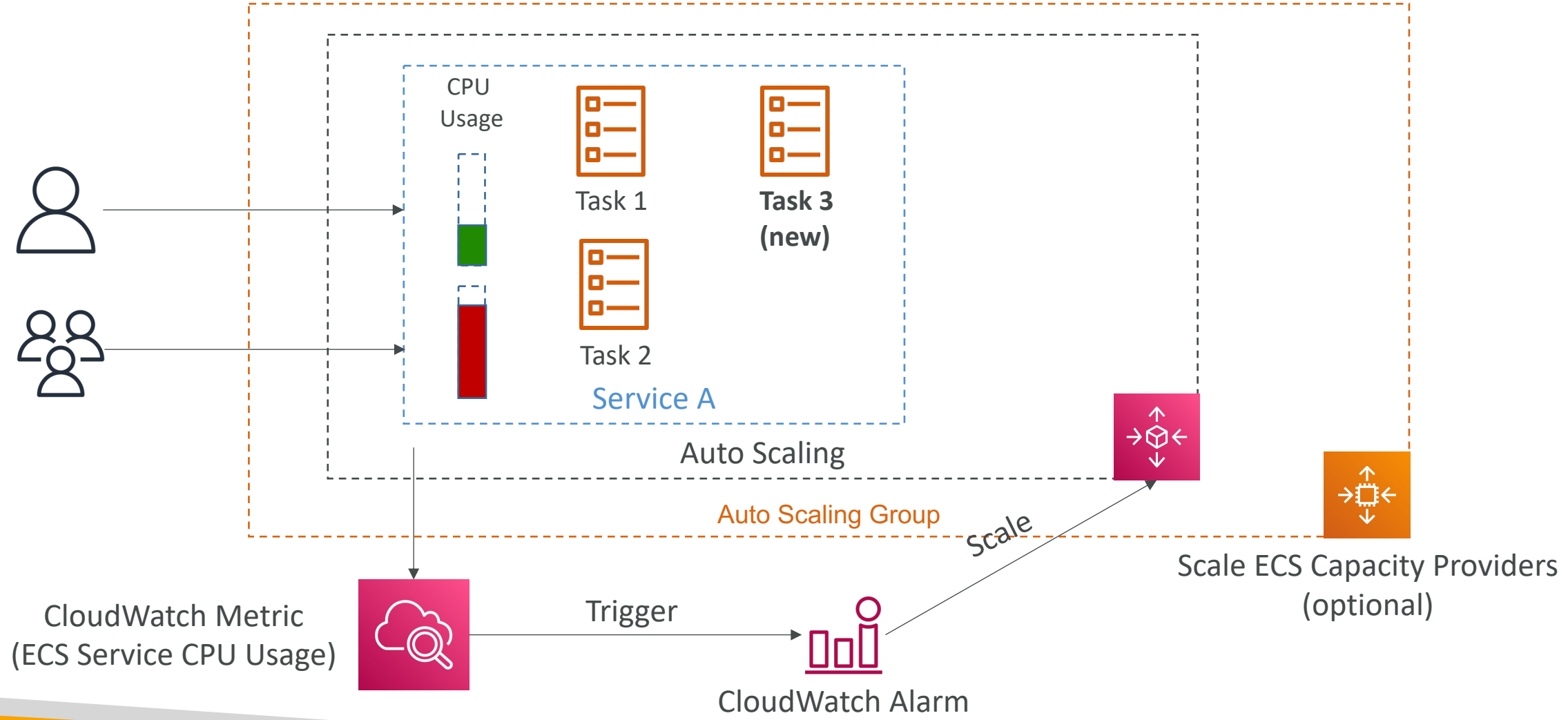
- Auto Scaling Group Scaling
  - Scale your ASG based on CPU Utilization
  - Add EC2 instances over time

- ECS Cluster Capacity Provider
  - Used to automatically provision and scale the infrastructure for your ECS Tasks
  - Capacity Provider paired with an Auto Scaling Group
  - Add EC2 Instances when you're missing capacity (CPU, RAM…)

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

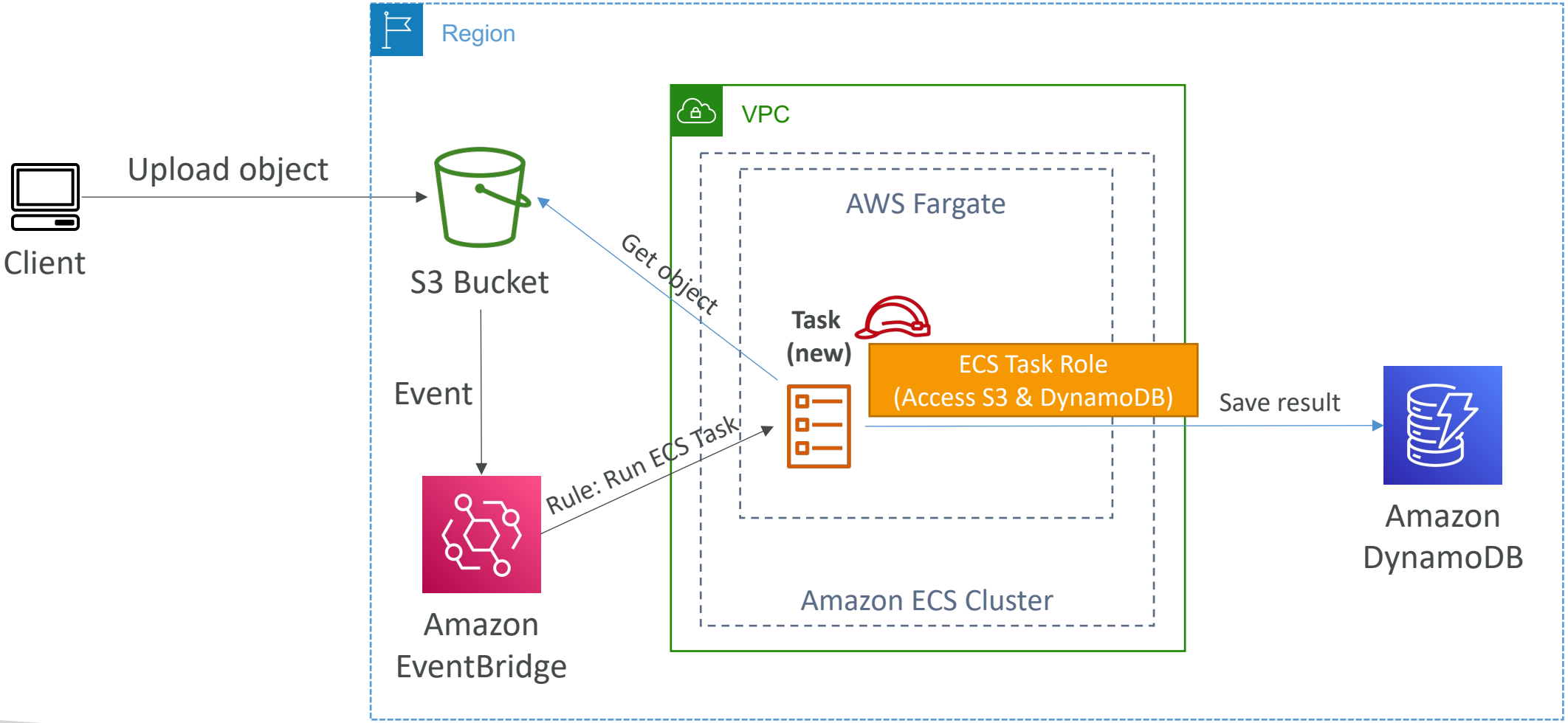
ECS Scaling – Service CPU Usage Example



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

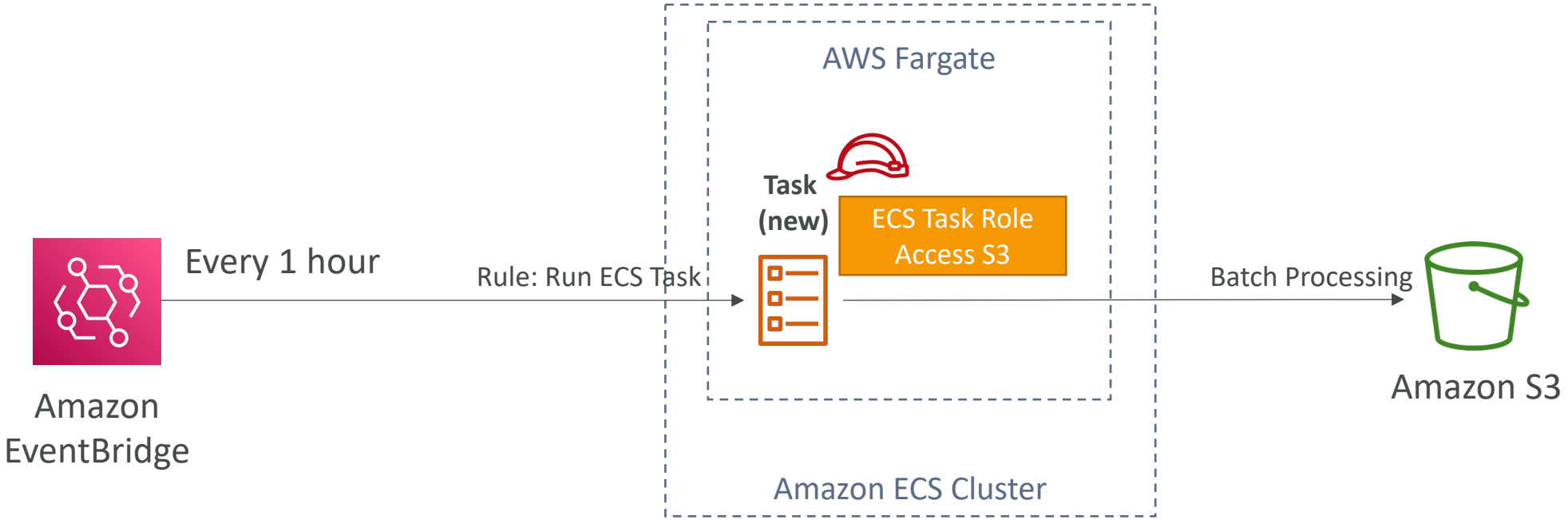
ECS tasks invoked by Event Bridge



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

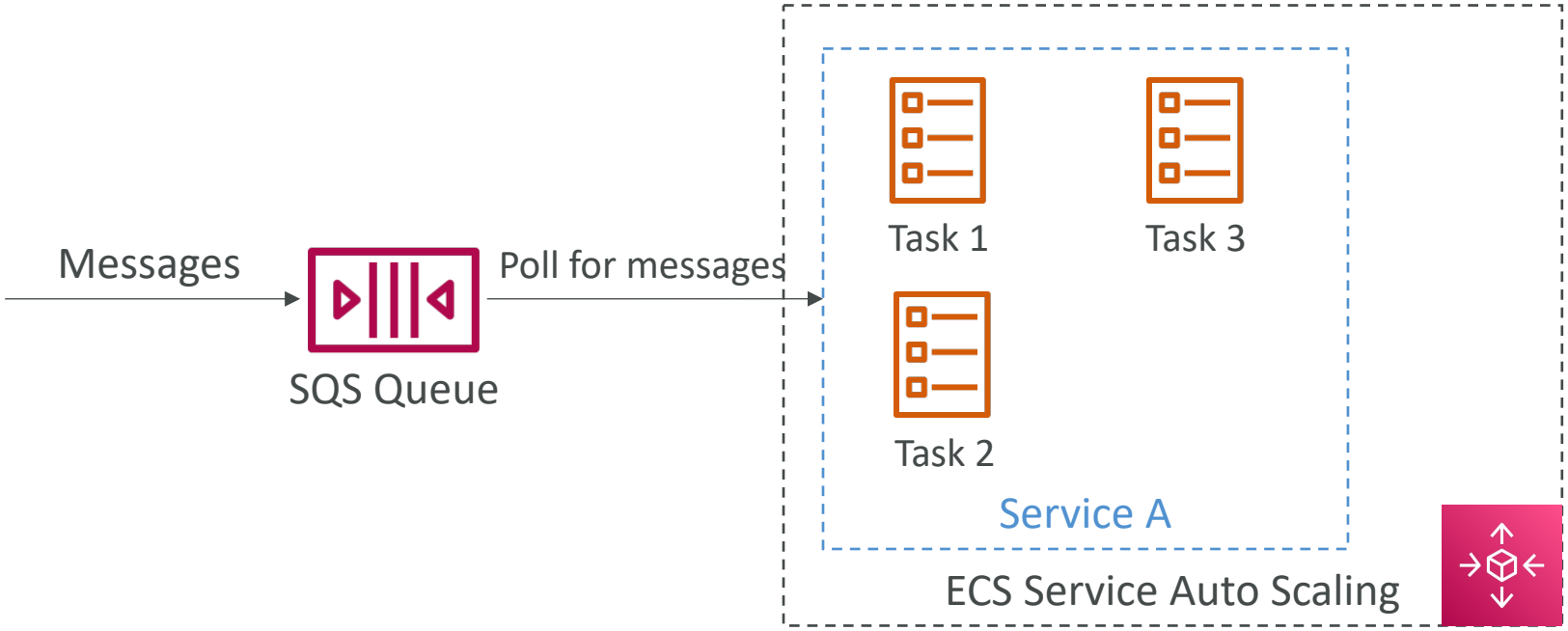
ECS tasks invoked by Event Bridge Schedule



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

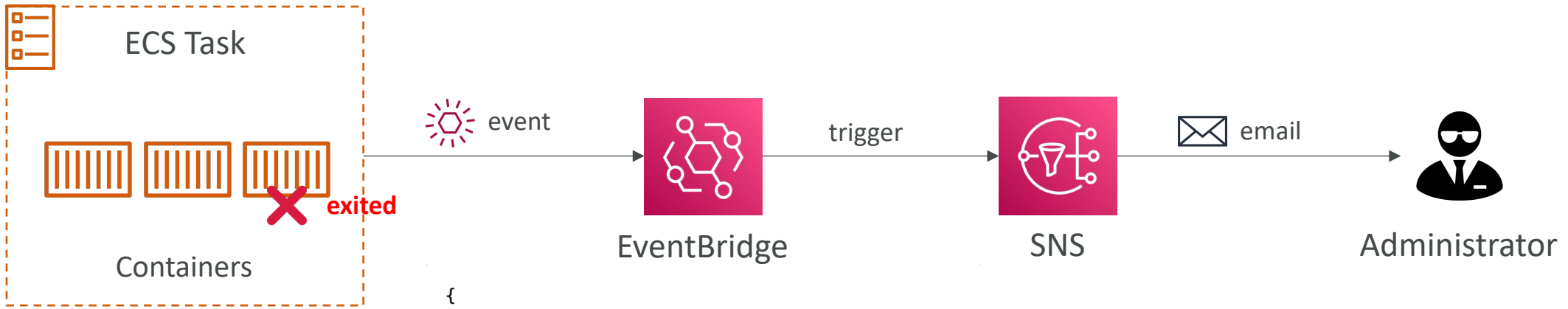
ECS – SQS Queue Example



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

ECS – Intercept Stopped Tasks using EventBridge




```
{
    "source": [
        "aws.ecs"
    ],
    "detail-type": [
        "ECS Task State Change"
    ],
    "detail": {
        "lastStatus": [
            "STOPPED"
        ],
        "stoppedReason": [
            "Essential container in task exited"
        ]
    }
}
```

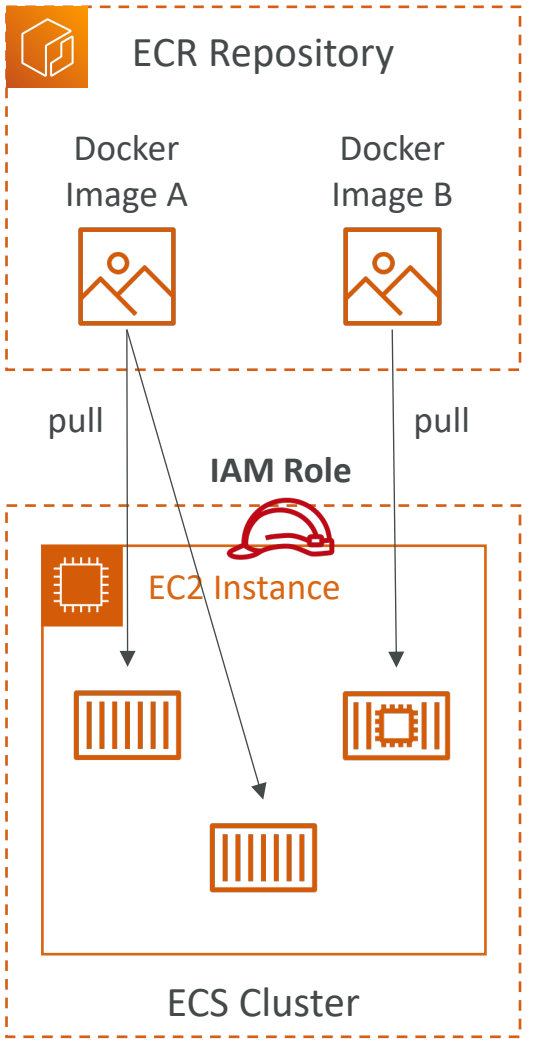
Event Pattern

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon ECR

- ECR = Elastic Container Registry
- Store and manage Docker images on AWS
- **Private** and **Public** repository (**Amazon ECR Public Gallery** https://gallery.ecr.aws)
- Fully integrated with ECS, backed by Amazon S3
- Access is controlled through IAM (permission errors => policy)
- Supports image vulnerability scanning, versioning, image tags, image lifecycle, …



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon EKS Overview



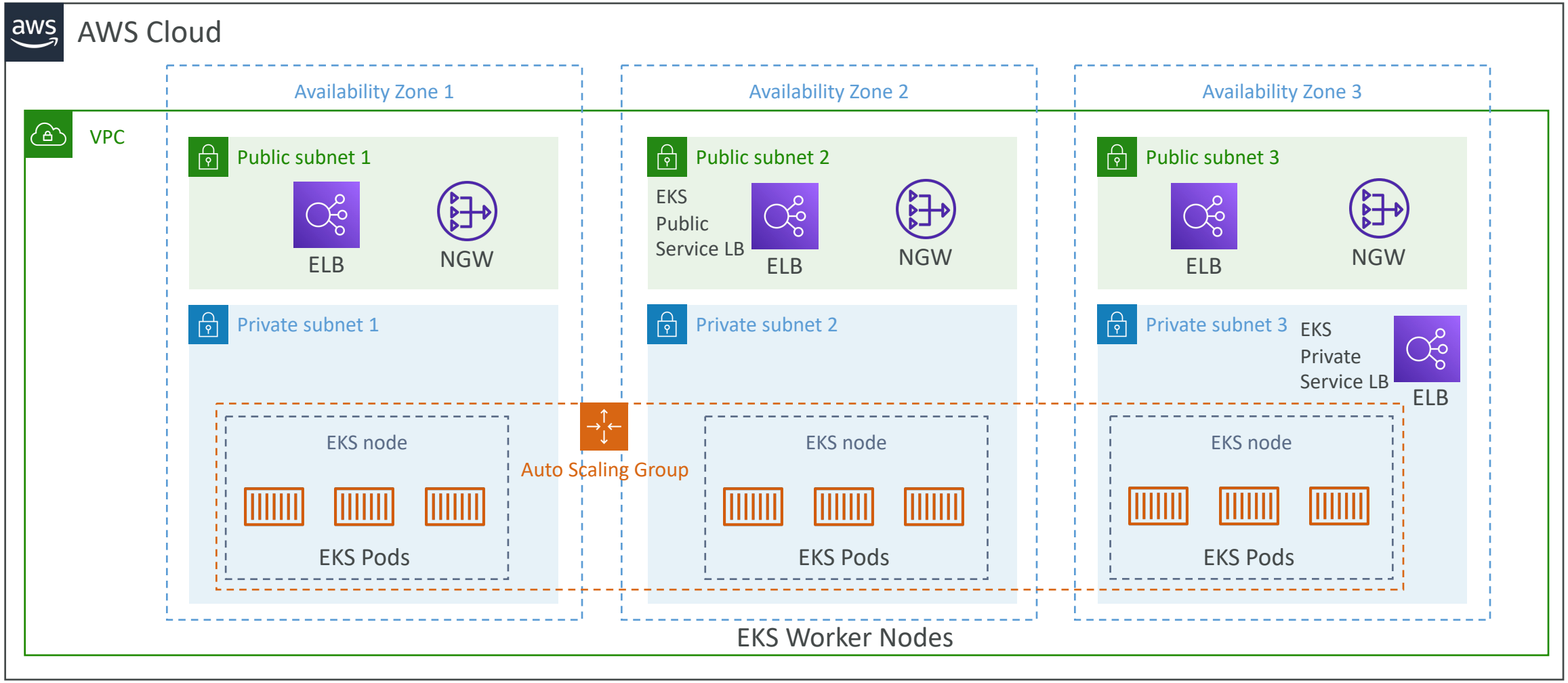


- Amazon EKS = Amazon Elastic **Kubernetes** Service
- It is a way to launch **managed Kubernetes clusters on AWS**
- Kubernetes is an **open-source system** for automatic deployment, scaling and management of containerized (usually Docker) application
- It's an alternative to ECS, similar goal but different API
- EKS supports **EC2** if you want to deploy worker nodes or **Fargate** to deploy serverless containers
- **Use case:** if your company is already using Kubernetes on-premises or in another cloud, and wants to migrate to AWS using Kubernetes
- **Kubernetes is cloud-agnostic** (can be used in any cloud – Azure, GCP…)
- For multiple regions, deploy one EKS cluster per region
- Collect logs and metrics using **CloudWatch Container Insights**

© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon EKS - Diagram



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon EKS – Node Types

- Managed Node Groups
  - Creates and manages Nodes (EC2 instances) for you
  - Nodes are part of an ASG managed by EKS
  - Supports On-Demand or Spot Instances
- Self-Managed Nodes
  - Nodes created by you and registered to the EKS cluster and managed by an ASG
  - You can use prebuilt AMI - Amazon EKS Optimized AMI
  - Supports On-Demand or Spot Instances
- AWS Fargate
  - No maintenance required; no nodes managed

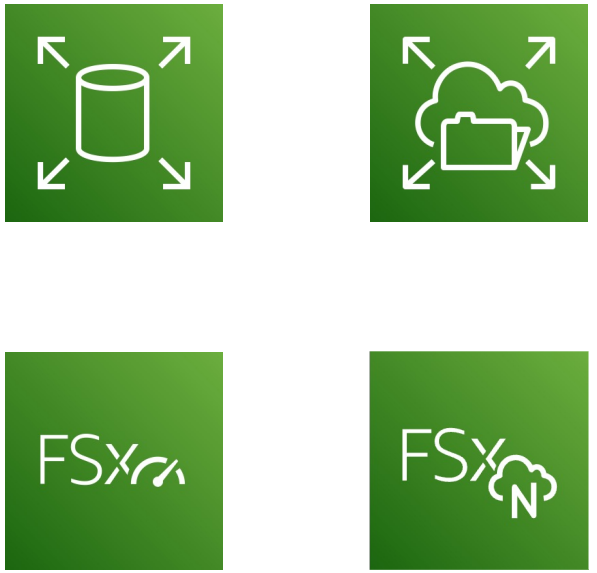
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

Amazon EKS – Data Volumes

- Need to specify **StorageClass** manifest on your EKS cluster
- Leverages a **Container Storage Interface (CSI)** compliant driver

- Support for…
- Amazon EBS
- Amazon EFS (works with Fargate)
- Amazon FSx for Lustre
- Amazon FSx for NetApp ONTAP



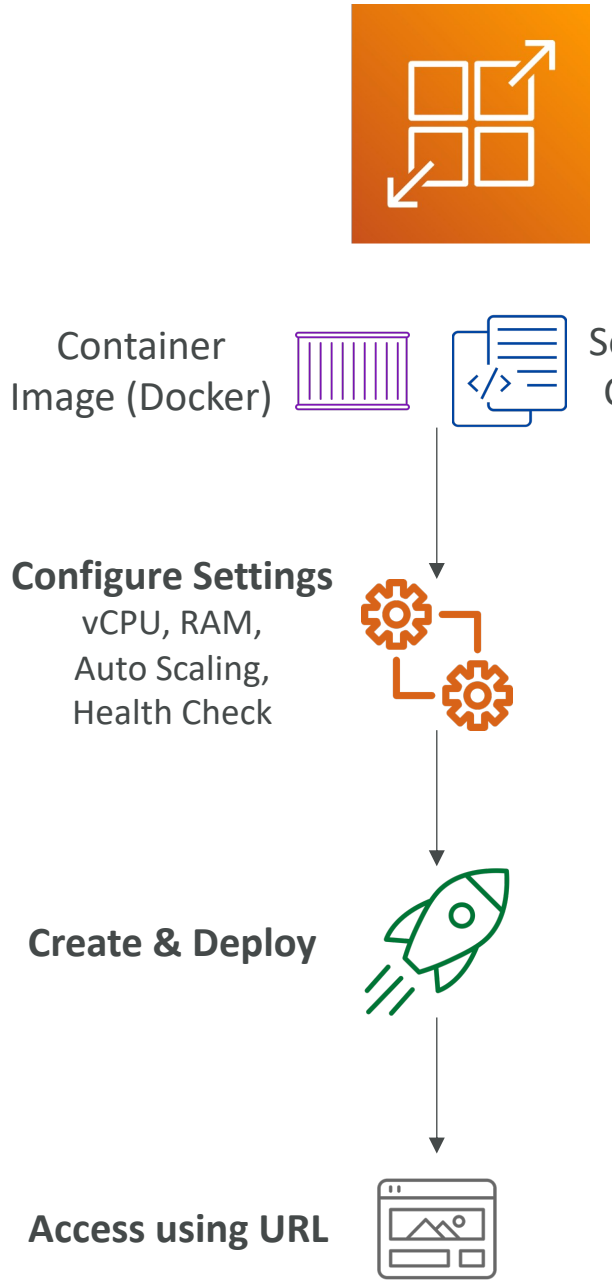
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS App Runner

- Fully managed service that makes it easy to deploy web applications and APIs at scale
- No infrastructure experience required
- Start with your source code or container image
- Automatically builds and deploy the web app
- Automatic scaling, highly available, load balancer, encryption
- VPC access support
- Connect to database, cache, and message queue services

- Use cases: web apps, APIs, microservices, rapid production deployments



© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

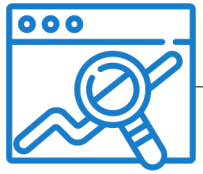
AWS App2Container (A2C)

- CLI tool for migrating and modernizing **Java** and **.NET** web apps into Docker Containers
- **Lift-and-shift** your apps running in on-premises bare metal, virtual machines, or in any Cloud to AWS
- Accelerate modernization, no code changes, migrate legacy apps…
- Generates CloudFormation templates (compute, network…)
- Register generated Docker containers to ECR
- Deploy to ECS, EKS, or App Runner
- Supports pre-built CI/CD pipelines

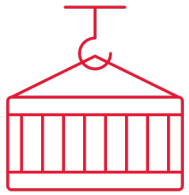
© Stephane Maarek

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

AWS App2Container (A2C)



Discover & Analyze
create app inventory and analyze runtime dependencies



Extract & Containerize
extract an app with dependencies and create a Docker image



Create Deployment Artifacts
generate ECS Task and EKS Pod definitions, and create CI/CD pipelines, and other infrastructure



CloudFormation Template



Deploy to AWS
store Docker image in ECR, and deploy to ECS, EKS, or App Runner



Amazon ECR
(store image)



Amazon ECS
(deploy)



Amazon EKS
(deploy)



App Runner
(deploy)

NOT FOR DISTRIBUTION © Stephane Maarek www.datacumulus.com

© Stephane Maarek